IOI Global vs Position-Aware Node Pruning

Finalized 500-epoch ABBA IOI runs · GPT-2 Small · hard-concrete masks · seed 42

Both circuits retained 100% IOI pairwise accuracy. The global baseline patched 52.7459% of global component paths to the corrupt stream. The position-aware circuit patched 79.7601% of component-position paths, while 18.6662% of physical components were patched at every logical position. These are causal activation-patching statistics, not model-compression statistics.

Meaning of a closed gate: `a_used = g × a_clean + (1 − g) × a_corrupt`. An open gate uses the clean activation; a closed gate substitutes the aligned corrupt-stream activation. Closed gates do not zero activations, remove weights, or skip computation. Both clean and corrupt streams are required by the current implementation.

Final test performance

Metric	Global baseline	Position-aware
Accuracy	100.0000%	100.0000%
KL divergence	0.1576847	0.1960384
Mean correct-minus-wrong logit difference	5.8987041	6.0766717
Top-1 next-token agreement with GPT-2	91.1469%	87.3239%
Soft faithfulness	1.7040985	1.7555122

Final hierarchy-aware activation-patching masks

Component type	Baseline global paths patched	Position-aware slots patched	Position-aware patched at all positions
Attention blocks	0 / 12 · 0.0000%	39 / 96 · 40.6250%	0 / 12 · 0.0000%
MLP blocks/layers	0 / 12 · 0.0000%	46 / 96 · 47.9167%	0 / 12 · 0.0000%
Attention heads	65 / 144 · 45.1389%	889 / 1,152 · 77.1701%	39 / 144 · 27.0833%
Attention neurons	5,322 / 9,216 · 57.7474%	61,196 / 73,728 · 83.0024%	3,654 / 9,216 · 39.6484%
MLP hidden neurons	21,042 / 36,864 · 57.0801%	238,023 / 294,912 · 80.7098%	6,235 / 36,864 · 16.9135%
MLP output dimensions	2,826 / 9,216 · 30.6641%	53,712 / 73,728 · 72.8516%	425 / 9,216 · 4.6115%
All gate types	**29,255 / 55,464 · 52.7459%**	**353,905 / 443,712 · 79.7601%**	**10,353 / 55,464 · 18.6662%**

Position-aware paths patched by logical section

Mask / section	Attn blocks	MLP blocks	Heads	Attn neurons	MLP hidden	MLP output
Baseline global	0.0000%	0.0000%	45.1389%	57.7474%	57.0801%	30.6641%
Prefix	100.0000%	100.0000%	100.0000%	100.0000%	100.0000%	100.0000%
First name	41.6667%	8.3333%	79.1667%	83.5286%	50.1845%	28.1359%
Name connector	50.0000%	91.6667%	88.1944%	90.4297%	99.0207%	98.5894%
Second name	41.6667%	25.0000%	77.7778%	83.4852%	71.4383%	65.4622%
Shared context	8.3333%	50.0000%	66.6667%	72.8950%	90.5355%	85.0477%
Repeated subject	33.3333%	8.3333%	70.1389%	82.0313%	66.7318%	52.6367%
Final action	25.0000%	50.0000%	79.1667%	84.8199%	90.8474%	84.1580%
Prediction relation	25.0000%	50.0000%	56.2500%	66.8294%	76.9206%	68.7826%

Runtime and extraction interpretation

Property of the current code	Global baseline	Position-aware
Requires clean stream	Yes	Yes
Requires corrupt stream	Yes	Yes
Weights physically removed	0%	0%
Computation skipped by a closed gate	0%	0%
Runnable as a standalone extracted model	No	No
Mask granularity	One decision per component	One decision per component and logical section

Parent closure was propagated before these mask counts were computed. A closed attention-block gate makes its head and attention-neuron paths effectively use the corrupt stream; a closed head does the same for its attention neurons; a closed MLP-block gate controls its hidden and output paths. For position-aware masks this is enforced independently in every section. Full-transformer- layer and embedding gating were disabled. Position-slot patching is an unweighted logical-section statistic, not literal parameter removal, model compression, or token-weighted FLOP reduction.